

## INTRODUCTION TO STATISTICAL METHODS

# Hypothesis Testing

Null and alternative hypotheses, significance and p-values, Type I/II errors, the Z-test and the t-test.

Companion reader · Read beside the slides

**How to use this companion.** Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

## 1 The logic of hypothesis testing

A **hypothesis test** asks whether the data give enough evidence to reject a default claim. We set up two hypotheses: the **null hypothesis**  $H_0$  (the “nothing interesting” claim, e.g. “the mean is 50”), and the **alternative**  $H_1$  (what we suspect, e.g. “the mean is *not* 50”). We assume  $H_0$  is true, then ask: how surprising is our data under that assumption? If very surprising, we reject  $H_0$ .

### The intuition

It is a courtroom.  $H_0$  is “innocent” (presumed true); the data are the evidence. We only convict (reject  $H_0$ ) if the evidence is strong *beyond reasonable doubt* — the doubt level is the significance  $\alpha$ .

## 2 Significance, p-values and errors

The **significance level**  $\alpha$  (often 0.05) is the risk we accept of rejecting a true  $H_0$ . The **p-value** is the probability, *if*  $H_0$  is true, of data at least as extreme as ours: small  $p \Rightarrow$  reject. Two mistakes are possible: a **Type I error** (reject a true  $H_0$  — a false alarm, probability  $\alpha$ ) and a **Type II error** (fail to reject a false  $H_0$  — a miss, probability  $\beta$ );  $1 - \beta$  is the **power**.

## 3 The Z-test for a mean

When the population standard deviation  $\sigma$  is known and the sample is large, standardise the sample mean:  $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ . The denominator  $\sigma/\sqrt{n}$  is the **standard error**. Compare  $Z$  to the critical values ( $\pm 1.96$  for a two-tailed test at  $\alpha = 0.05$ ).

### Worked example: a two-tailed Z-test

Claim  $H_0 : \mu = 50$ . Sample of  $n = 64$  has mean  $\bar{x} = 52$ ; known  $\sigma = 8$ . Standard error  $= \frac{8}{\sqrt{64}} = \frac{8}{8} = 1$ . Test statistic  $Z = \frac{52 - 50}{1} = \mathbf{2.0}$ . Since  $|2.0| > 1.96$ , we **reject**  $H_0$  at  $\alpha = 0.05$ . The p-value is  $2P(Z > 2) = 2(0.0228) = \mathbf{0.0455} < 0.05$  — consistent with

rejection.

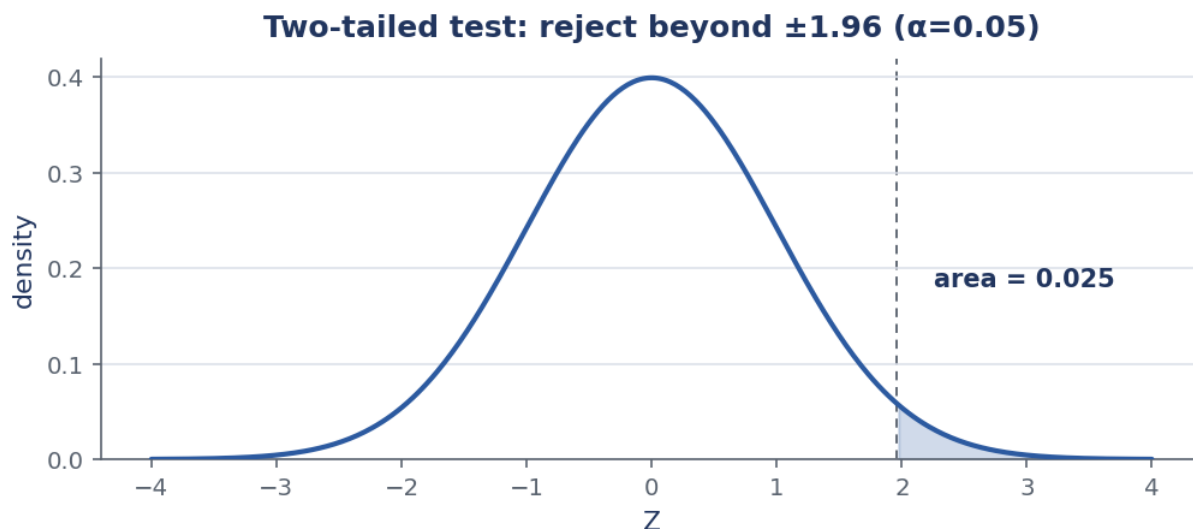


Figure 1: At  $\alpha = 0.05$  two-tailed, the rejection region is the shaded tails beyond  $\pm 1.96$ ; our  $Z = 2.0$  falls inside it, so we reject  $H_0$ .

## 4 The t-test: when $\sigma$ is unknown

In practice  $\sigma$  is rarely known. We estimate it with the sample standard deviation  $s$  and use the **t-statistic**  $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ , compared against the **t-distribution** with  $n - 1$  **degrees of freedom**. For small samples the t-distribution has fatter tails (more caution); for large  $n$  it approaches the normal. The same idea extends to the **difference of two means** (two-sample t-test).

### ✓ Key takeaway

A hypothesis test weighs evidence against a null  $H_0$  at significance  $\alpha$ ; the **p-value** is how surprising the data are under  $H_0$ . The **Z-test** uses  $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$  (worked:  $Z = 2.0$ ,  $p = 0.0455 \Rightarrow$  reject); the **t-test** replaces  $\sigma$  with  $s$  and uses  $n - 1$  degrees of freedom. Beware **Type I** (false alarm,  $\alpha$ ) and **Type II** (miss,  $\beta$ ) errors.